Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 1 – Complex Numbers

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Express $3 + 4i$ in modulus-argument form. **[3]**

2. Find the modulus and argument of $1 - i$. **[3]**

3. Calculate $(2 + 3i)(1 - i)$. **[3]**

4. Find the real and imaginary parts of $\dfrac{1 + 2i}{3 - i}$. **[4]**

5. Find $z\bar{z}$ if $z = 5 - 2i$. **[2]**

6. Solve $z^2 + 4 = 0$. **[3]**

7. Find the complex conjugate of $3 - 7i$. **[1]**

8. Express $\sqrt{3} + i$ in modulus-argument form. **[3]**

9. Calculate $(1 + i)^4$. **[4]**

10. Find $\dfrac{2}{i}$. **[2]**

11. If $z = x + yi$, find $z + \bar{z}$ and $z - \bar{z}$. **[3]**

12. Solve $z^2 - 2z + 5 = 0$. **[4]**

13. Find the modulus of $\dfrac{3+4i}{1+2i}$. **[3]**

14. Express $5(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$ in the form $a + bi$. **[3]**

15. Find i^{15}. **[2]**

Section B – Hard (Q16–25)

16. Find the square roots of $3 + 4i$. **[6]**

17. Solve $z^2 + (2-i)z + (3+i) = 0$. **[6]**

18. Given $z = 2 + i$, find z^3 in the form $a + bi$. **[4]**

19. Prove that $z\bar{z} = |z|^2$. **[3]**

20. Find the values of x and y if $(x + yi)(2 - i) = 5 + 5i$. **[4]**

21. Express $\dfrac{1+i}{1-i}$ in the form $re^{i\theta}$. **[4]**

22. Find all complex numbers z such that $z^2 = \bar{z}$. **[5]**

23. If $z = cis\theta$, show that $z^n + \frac{1}{z^n} = 2\cos n\theta$. **[5]**

24. Solve $z^4 + 1 = 0$ and express the roots in modulus-argument form. **[6]**

25. Given $|z - 1| = |z + i|$, describe the locus of z and find its Cartesian equation. **[5]**

End of Worksheet • *Show all working clearly.*